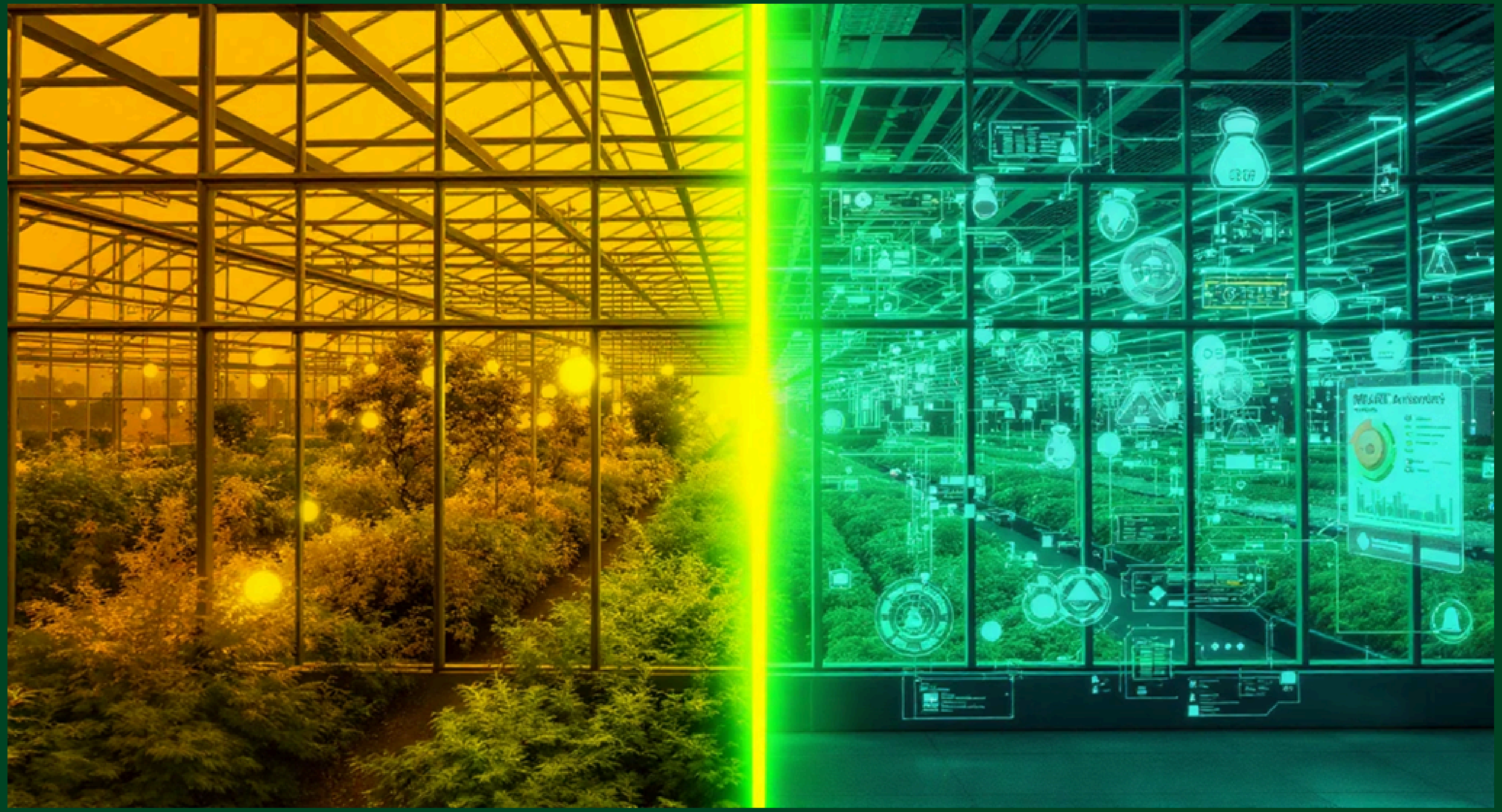
SMART GREENHOUSE AUTOMATION SYSTEM

IoT-based Agricultural Automation

PRESENTED BY : MONISHA C

INTODUCTION

- Modern agriculture faces critical challenges in maximizing yield while minimizing resource use.
- Traditional greenhouses rely on manual monitoring. Smart solutions use data to make decisions.
- Our system integrates sensors, controls, and analytics for optimal growing conditions.



OBJECTIVES

- Automate irrigation, ventilation, and lighting systems.
- Monitor environmental factors in real time.
- Improve plant health and reduce resource use.
- Enable remote control and alerts via mobile or web.

KEY FEATURES

- Automatic irrigation based on soil moisture
- Real-time environment monitoring
- Light control for optimized growth
- Data display via OLED/LCD or mobile app
- Energy-efficient and low-cost design

FUTURE ENHANCEMENTS

- Camera for plant image monitoring
- Machine learning for disease prediction
- Solar panel integration
- Mobile app alerts and notifications

PROBLEM STATEMENT

- Traditional greenhouses require manual monitoring and control.
- This leads to water and energy wastage.
- Plants can suffer due to delayed human intervention.

ESSENTIAL SENSORS

TEMPERATURE

LM35 and DHT11/22 sensors monitor ambient and soil temperatures.

MOISTURE

SEN0114 sensors track soil moisture levels for optimal watering

LIGHT

Photoresistors and PAR sensors measure light intensity and spectrum

AIR QUALITY

CO_2, humidity, and TVOC sensors ensure ideal atmospheric conditions.

SYSTEM OVERVIEW

Sensor Network

Distributed sensors continuously monitor all environmental parameters

Data Collection

Real-time information gathered and processed by central controller

Automated Response

System adjusts conditions based on data and predefined parameters

Cloud Platform

Access monitoring and controls via web interface or mobile app

SYSTEM BLOCK DIAGRAM

Sensor Network

Environmental data collection from multiple sensor types.

Control Unit

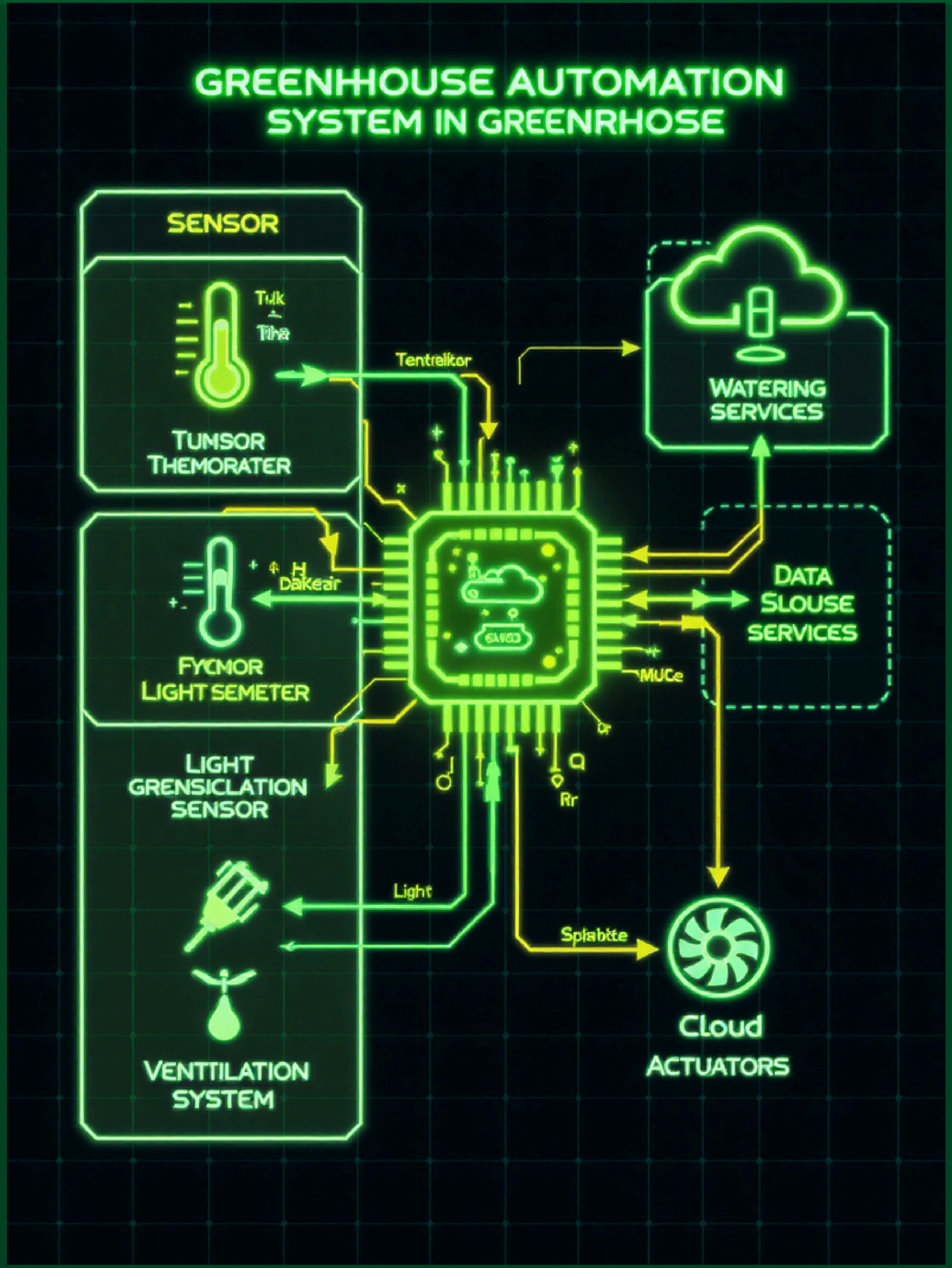
Central processor analyzes inputs and executes control algorithms.

Data Transmission

Wireless/wired communication to cloud and local interfaces.

Output Controls

Precise adjustment of environmental factors via actuators.



TECHNOLOGIES USED

- Arduino / ESP8266 / ESP32
- DHT11 / DHT22 (Temperature & Humidity sensor)
- Soil Moisture Sensor, LDR (Light sensor)
- Relay module for automation
- ThingSpeak / Blynk / Firebase for cloud integration

KEY HARDWARE COMPONENTS

Environmental Sensors

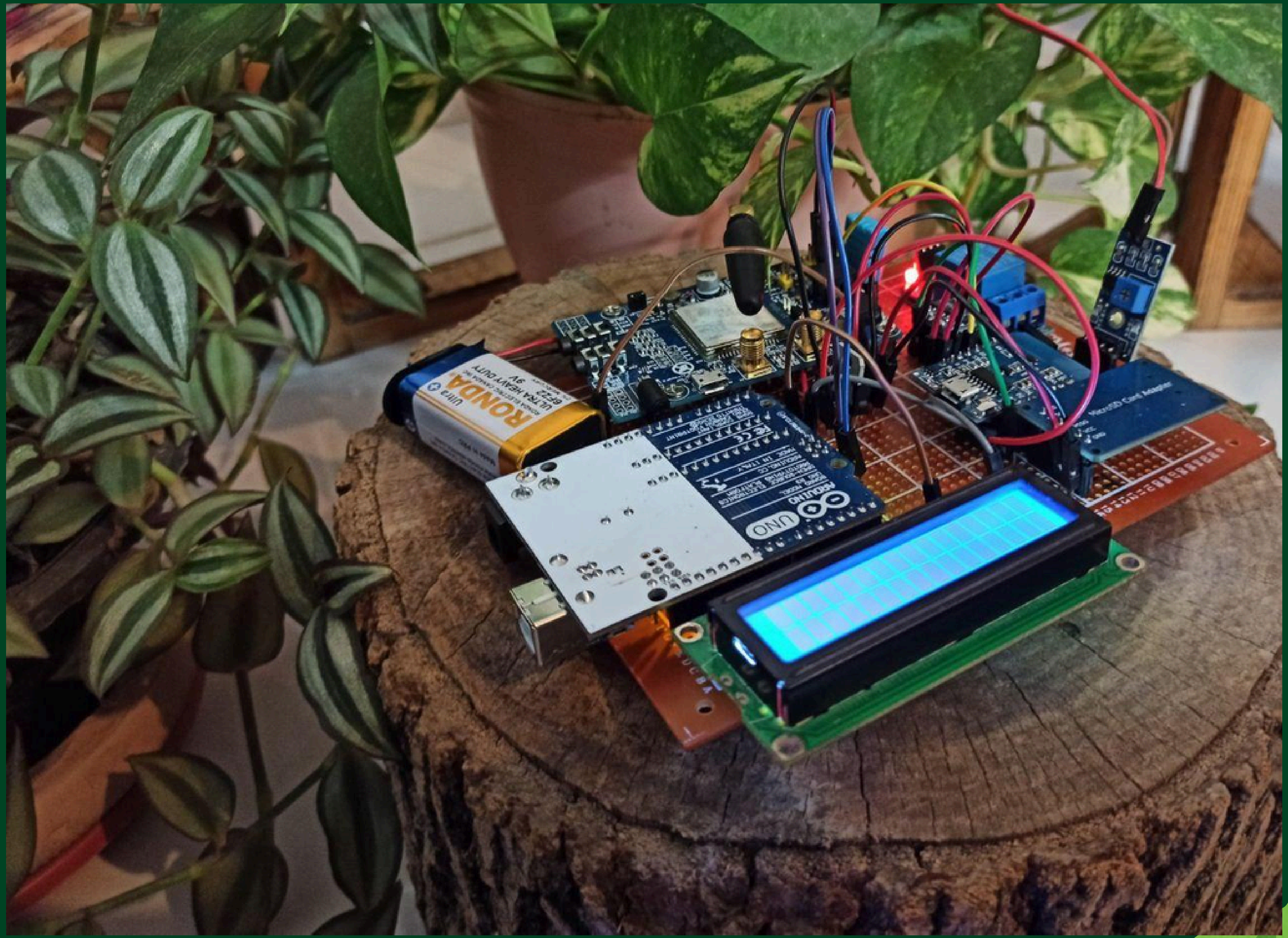
Monitor CO_2, temperature, humidity, light, soil moisture, and TVOCs.

Control Mechanisms

Fans, pumps, valves, and actuators regulate the growing environment.

Processing Units

Arduino microcontrollers manage system operations.



CONCLUSION

www.reallygreatsite.com

- The Smart Greenhouse Automation System offers an innovative solution for modern agriculture.
- It successfully automates key tasks like irrigation, ventilation, and lighting using real-time sensor data.
- Integration with IoT platforms enables farmers to monitor and control the greenhouse remotely.
- This system ensures optimal growing conditions, reduces human effort, and saves valuable resources like water and electricity.
- The project is cost-effective, scalable, and suitable for both small-scale and large-scale farming.
- With future enhancements like AI and solar power, it can play a major role in building sustainable and smart farming practices.
- Overall, this project is a step toward efficient, technology-driven agriculture for a better tomorrow.

THANK YOU